



Luxury hotel in Space

The Aurora Station, the first luxury hotel in space, will launch in 2021 and house four guests, two crew members. <https://digit.in/hotelinspc>



Listening to an Interstellar visitor

An outback radio telescope in Western Australia has been listening in on a cigar-shaped interstellar object. <https://digit.in/radiotele>

subsequently implanting those eggs into surrogate mothers. After 57 of these attempts, only 7 instances of pregnancy was achieved, out of which only one survived till delivery. The 4.5 pounds goat didn't survive for long as she had a gigantic extra lobe in her lungs that made it impossible to breathe.

This was almost 15 years ago. The method involved has been improved to a vast extent and SCNT, in this manner, is way more reliable now. The technol-



Passenger Pigeons were abundant enough for their flocks to block the sun when they flew by

ogy is no longer a high-risk endeavor. Researchers are now able to induce induce adult animal cells to return to an embryo-like state. These cells can then be made to develop into any type of cell – including eggs or sperm – which can then be further manipulated to develop into full-fledged embryos.

GENOME EDITING

In the early 1800s, nobody would have thought that the Passenger Pigeon, a species abundant enough to block out the sun when flocks flew by, would become extinct a hundred years down the line. Yet, decades of relentless hunting later, the last passenger pigeon would perish in captivity in 1914, at the Cincinnati Zoo.

Environmentalist and author Stewart Brand has led the effort to bring back the species through a project under Revive and Restore (revive&restore), an organisation that aims to apply new genetic technology to solve biodiversity problems. The

project, dubbed the Great Passenger Pigeon Comeback, began in 2012. The process that they are following also sets the outline for bringing back quite a few other extinct bird species.

To begin with, the genomes of the Passenger Pigeon and the Band Tailed pigeon, its closest living relative, are compared. While the genome for the Band Tailed pigeon is readily available, that of the Passenger Pigeon has to be generated by comparing sequences

from its remains. After the comparison, the areas where they aren't same are identified. Primordial germ cells (PGCs) are extracted from a band tailed pigeon which is then edited using CRISPR to write in the 'passenger pigeon' DNA. These edited PGCs are then injected into a rock pigeon where they migrate to the reproductive organs. After successful breed-

ing, the first generation of passenger pigeons will be born all over again.

Currently, the project has successfully completed and sequenced the entire genome for the Band tailed pigeon and have sequenced DNA from 37 passenger pigeons. Joined by the the Center for Genome Architecture at Rice University's Baylor College of Medicine in 2017, the project has begun to edit Domestic Rock Pigeon genome to test the feasibility of the method. The next step is to raise enough money for the germ line transmission. The project aims to achieve de-extinction by 2020 and release a flock of 10,000 Passenger Pigeons into the wild by 2030.

THE BENEFITS

While we've dealt with the moral question behind de-extinction, they are far reaching biological and ecological implications of bringing back a species that does not exist out there in the wild anymore. The impact of the revived

species, as an invading 'alien' species, with respect to the currently balance in biodiversity, has been questioned significantly. Additionally, doubts have been raised on the ability to sustain the revived species. The first ones brought back to life would be raised by parents from a different species, which might interfere in their expected natural lifecycle or change it altogether. The biologists working on these projects have answers to these questions. Take the passenger pigeon for instance - most of its behavior is encoded into its genes, thus nullifying the effect that being raised by other species might have on it.

The other side, things look good. Bringing back a species like the passenger pigeon might actually go a long way in protecting the natural biodiversity in the region it originally dwelt in. Australia's southern gastric-brooding frog possessed a unique ability to incubate its eggs in its stomach, effectively turning it into a uterus. This ability made it a medical marvel to be studied before it went extinct due to widespread fungicide abuse. Bringing back a species like this could help us



The Passenger Pigeon (*Ectopistes migratorius*)

gain an understanding of gastrointestinal enzymes and how we can control them. Even bringing back the Woolly Mammoth would restore the diversity of its original habitat to a great extent. The benefits of bringing back extinct species clearly outweighs the probable disadvantages. The important question is, once we bring them back, will we let them down again? Let's hope not. 